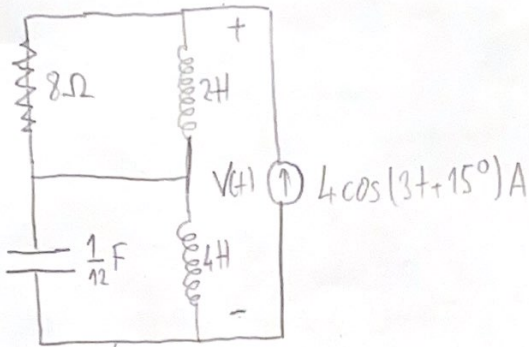


# EE 201 - Assignment #4

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1)



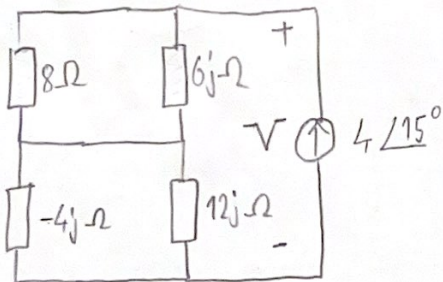
\* We first need to find impedances of each circuit element.

$$Z_C = \frac{1}{j\omega C} \quad Z_L = j\omega L$$

$$i(t) = 4 \cos(3t + 15^\circ) \text{ A}$$

$$I = 4 \angle 15^\circ \quad \omega = 3 \text{ rad/s}$$

$$Z_{2H} = 6j \Omega, \quad Z_{4H} = 12j \Omega, \quad Z_{\frac{1}{12}F} = -4j \Omega, \quad Z_{8\Omega} = 8 \Omega$$



$$Z_{eq} = (8 \parallel 6j) + (-4j \parallel 12j)$$

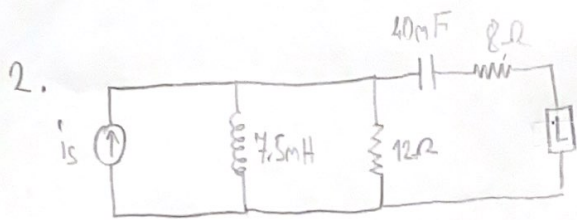
$$= \left( \frac{3j}{24j} + \frac{4}{24j} \right)^{-1} + \left( \frac{-3}{12j} + \frac{1}{12j} \right)^{-1}$$

$$= \frac{24j}{3j+4} - 6j = \frac{+18}{3j+4} \Omega = \frac{18(4-3j)}{25}$$

$$Z_{eq} = \frac{18}{25} \cdot 5 e^{j \tan^{-1}(-\frac{3}{4})} = \frac{18}{5} \angle -36.87^\circ$$

$$V = I Z_{eq} = 4 \angle 15^\circ \cdot \frac{18}{5} \angle -36.87^\circ = \frac{72}{5} \angle -21.87^\circ$$

$$V(t) = 14.4 \cos(3t - 21.87^\circ) \text{ V}$$



$$i_s = 5 \cos(40t) \text{ A}$$

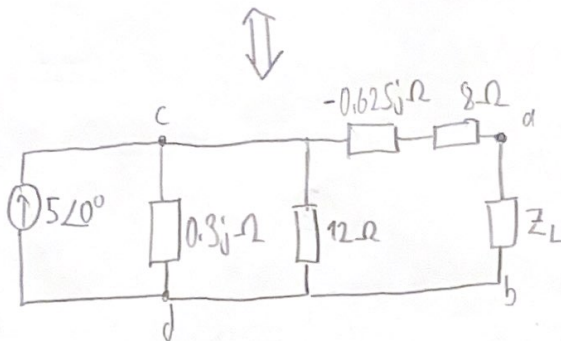
$$\omega = 40$$

$$I = 5 \angle 0^\circ$$

$$Z_{7.5\text{mH}} = j 7.5 \times 10^{-3} \times 40 = 0.3j \Omega$$

$$Z_{40\text{mF}} = \frac{1}{j \times 40 \times 40 \times 10^{-3}} = -0.625j \Omega$$

$$Z_{12\Omega} = 12 \Omega \quad Z_{8\Omega} = 8 \Omega$$

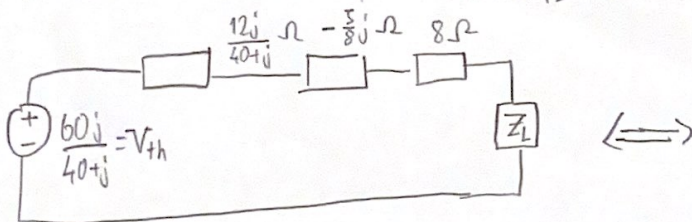


\* We need to make a series of source transformations in order to obtain an equivalent Thevenin circuit.

$$Z_{cd} = \left( \frac{1}{0.3j} + \frac{1}{12} \right)^{-1} = \left( \frac{40}{12j} + \frac{j}{12j} \right)^{-1} = \frac{12j}{40+j} \Omega$$

$$V_{th} = 5 \angle 0^\circ \cdot \frac{12j}{40+j} = \frac{60j}{40+j}$$

→ Then our circuit transforms into



$$Z_{th} = \frac{12j}{40+j} - \frac{5j}{8} + 8 = \frac{96j - 5j(40+j) + 8(320+8j)}{320+8j} = \frac{96j - 200j + 5 + 2560 + 64j}{320+8j}$$

$$Z_{th} = \frac{2565 - 40j}{320+8j} = 5 \left( \frac{513 - 8j}{320+8j} \right)$$

\*  $Z_L$  must be  $Z_{th}^*$  for maximum power transfer

$$Z_L = 5 \cdot \frac{513+8j}{320-8j} \approx 8 + 0.32j = 8 \angle 2.32^\circ$$